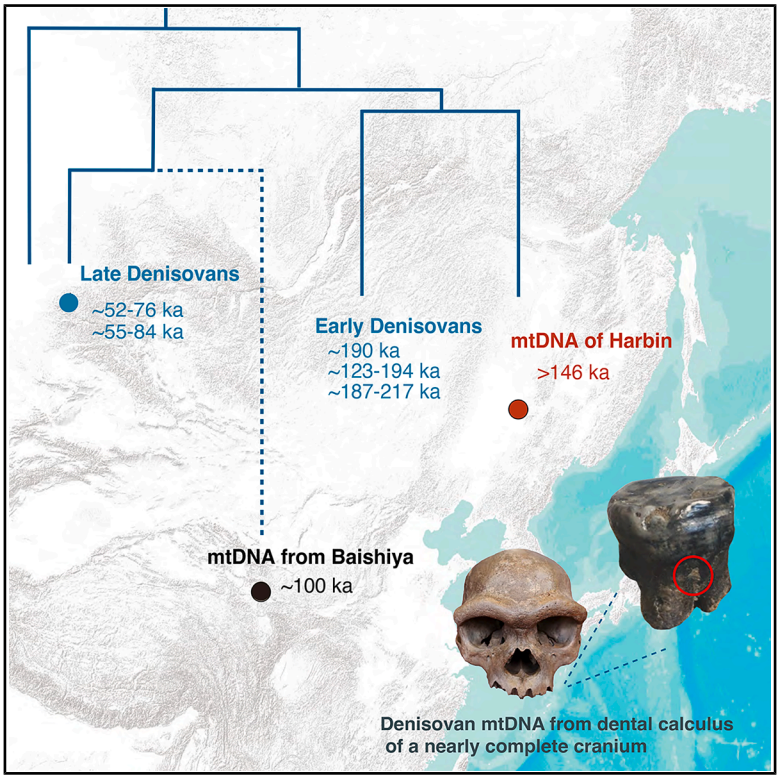
Cell

Short article

Denisovan mitochondrial DNA from dental calculus of the >146,000-year-old Harbin cranium

Graphical abstract



Highlights

- Host DNA was retrieved from the dental calculus of a Middle Pleistocene hominin
- The Harbin mtDNA (>146 ka) is linked to early Denisovan mtDNAs
- Denisovan mtDNA is directly connected to a nearly complete hominin cranium

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In brief

A mitochondrial genome from Middle Pleistocene hominin dental calculus from northeastern China clusters with mtDNAs of early Denisovan individuals in Siberia, linking a nearly complete cranium to Denisovans.

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Short article

Denisovan mitochondrial DNA from dental calculus of the >146,000-year-old Harbin cranium

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SUMMARY

Denisovans have yet to be directly associated with a hominin cranium, limiting our understanding of their morphology and geographical distribution. We have attempted to retrieve DNA from a nearly complete Middle Pleistocene cranium from Harbin (>146 ka), northeastern China. Although no DNA could be retrieved from a tooth or the petrous bone, mitochondrial DNA (mtDNA) could be isolated from dental calculus. The mtDNA falls within Denisovan mtDNA variation and is related to an mtDNA branch carried by early Denisovan individuals in southern Siberia, previously observed in Denisova Cave. This suggests that Denisovans inhabited a large geographical range in Asia in the Middle Pleistocene. The association of Denisovan mtDNA with the Harbin cranium allows a better understanding of the morphological relationships between Denisovans and other East Asian Middle Pleistocene fossils. Furthermore, the retrieval of host DNA from dental calculus opens new possibilities for genetic research on Middle Pleistocene hominins.

INTRODUCTION

More than 20 Middle Pleistocene (78–300 thousand years [ka] ago) sites have been identified in China.[1–6] With improvements in dating and morphological studies of fossils at these sites, our understanding of ancient populations in East and Southeast Asia during this period is greatly enhanced. Hominin populations from Middle Pleistocene East Asia show a high degree of morphological diversity after 300 ka.[7] However, in spite of this, no genetic contribution from archaic hominins other than Denisovans or Neanderthals has been identified in Paleolithic or present-day Asian populations. Sampling of ancient DNA from Middle Pleistocene sites in North Asia and Europe[8–13] has shown the power of genetic evidence to clarify the relationships of Middle Pleistocene humans to present-day populations. One discovery made possible by the retrieval of ancient DNA was the identification of an archaic human population from Denisova Cave in Siberia, referred to as Denisovans, that was distinct from Neanderthals.[8,9] However, the morphology of Denisovans remains largely unknown because Denisovan DNA recovered to date derives from fragmented bones and teeth that possess little morphological information.[14] Further, no direct DNA evidence linked to morphology has been found in present-day China, a region that Denisovan populations are likely to have inhabited.[8–10,14]

Denisovans coexisted with modern humans and are a sister group to Neanderthals,[8–10] with whom they shared ancestors around 400,000 years ago.[15] Studies of Denisovan DNA from fossils and indirectly from introgressed fragments in modern human genomes show that at least two Denisovan lineages have introgressed into modern humans.[9,10,15–20] Of these, one is more closely related to individuals from Denisova Cave than the other.[20] Although Denisovans are speculated to have once had a large geographical range throughout Asia, to date, the only molecular evidence of Denisovans in Asia is from teeth, fragmentary bones, and cave sediments from Denisova Cave in Siberia[8–13,21]; a mandible, rib, and cave sediments from Baishiya Cave on the Tibetan Plateau[4,22,23]; and a mandible from the Penghu Channel in Taiwan.[24] Furthermore, samples from a cave in Laos (Tam Ngu Hao 2) have been suggested to be Denisovan based on morphological analysis.[25]

Throughout this study, we refer to "Denisovans" as the 7 previously described individuals from Denisova Cave (*Denisova 2*, *Denisova 3*, *Denisova 4*, *Denisova 8*, *Denisova 19*, *Denisova 20*, and *Denisova 21*),[8–13] excluding other individuals and their genetic data due to a more distant genetic relationship (Sima de los Huesos),[26,27] ambiguous relationships to individuals from

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Figure 1. The Harbin cranium and geographic location of hominin specimens older than 100 ka where human DNA has been retrieved

(A) The Middle Pleistocene cranium recovered in Harbin, Heilongjiang Province, northeastern China. (B) Map of geographic locations and dates for some sites where human DNA from individuals older than 100 ka has been retrieved. Harbin ~146 ka (sampled in this study), Baishiya Karst Cave ~100 ka (sedimentary DNA), Denisova Cave 123–217 ka (bones from layers 14 and 15 of the East Chamber and layer 22.1 of the Main Chamber), Scladina Cave ~120 ka, and Hohlenstein-Stadel Cave ~124 ka. Circles indicate Denisovan mtDNA or Denisovan-like mtDNA; squares indicate Neanderthal mtDNA.

Denisova Cave (Tam Ngu Hao *2*), or a lack of associated skeletal fragments (Baishiya Cave sediment).

Confirming the presence of Denisovans can be challenging for several reasons. For example, it is unclear whether the specimen from Laos falls within Denisovan variation, as the proteomic analysis has not yielded genetic markers specific to Denisovans. Moreover, our understanding of Denisovan morphology is limited. Even when genetic material is available, identifying a specimen as a Denisovan is not straightforward. For example, the mitochondrial DNA (mtDNA) from 400,000-year-old individuals found at Sima de los Huesos falls outside the known Denisovan mitochondrial diversity yet is more similar to Denisovan mtDNA than to that of Neanderthals, whereas their nuclear DNA is more closely related to Neanderthals.[26,27]

A notable late Middle Pleistocene cranium has been recovered in Harbin, Heilongjiang Province, northeastern China and dated to a minimum age of 146 ka[2] (Figure 1A). This cranium has been classified as a new species, named *Homo longi*, based on its morphology.[2] However, its morphology has been suggested to be related to the Xiahe mandible,[1] which has been associated with Denisovans through proteomic analysis.[4] No genetic information has been retrieved from the Harbin specimen, presumably due to its age. In fact, human DNA has only been sequenced from skeletal remains from four sites that are older than 100 ka: individuals dating to ~400 ka from Sima de los Huesos[26,27]; two Neanderthals dating to ~120 ka from Hohlenstein-Stadel Cave in southwestern Germany[28] and Scladina Cave in Belgium[29]; bone fragments from Denisova Cave dating to 187–217,[13] 190,[30] and 123–194 ka[31] (Figure 1B).

RESULTS

Ancient DNA retrieval and DNA sequence determination

Our initial attempts to retrieve DNA from a tooth or petrous bone of the Harbin cranium failed (STAR Methods and Table S1). We then turned to dental calculus (calcified dental plaque), which can provide a protected environment for DNA preservation[32,33] due to its dense crystalline structure that resists degradation in various environments.[34] Indeed, recent studies[32,33,35,36] have retrieved endogenous ancient DNA (i.e., DNA from the host individual and oral microbiome) from dental calculus, whereas, to date, only DNA from the hosts' microbiome and pathogenic microorganisms has been retrieved from Paleolithic dental calculus.

To reduce the presence of present-day DNA, the Harbin dental calculus was treated with bleach (STAR Methods)[37] before DNA extraction. Then, we extracted DNA from two dental calculus samples (0.5 and 0.3 mg, Figure 2) and converted aliquots of the extracts into single-stranded libraries. The two libraries were enriched with modern human mtDNA probes,[38] and 142,981 and 155,085 fragments, respectively, were sequenced from the libraries. Fragments shorter than 30 base pairs (bp) or with a mapping quality lower than 25 were discarded, resulting in 3,665 and 2,047 unique fragments, respectively, mapping to human mitochondrial genomes. The frequencies of C to T substitutions in the 5′ ends of the DNA fragments in these two libraries were 2% and 25%. Elevated levels of these substitutions are due to cytosine deamination that accumulates over time and thus indicates the presence of ancient DNA molecules. C to T frequencies of less than 5% are considered evidence of little to no ancient DNA. Thus, we retained the data from the library with a higher C to T substitution frequency (i.e., the 0.3-mg sample), where we were able to recover human mtDNA to a total of 6.6-fold coverage, presumably including both ancient DNA and contaminating modern DNA.

To maximize the mtDNA fragments available for analysis, we generated additional libraries from the remaining DNA extracts, as well as re-extracts from the silica membrane and pellet from both samples (STAR Methods). Eight out of a total of 20 libraries prepared from the two samples had C to T substitution frequencies above 5% at the terminal ends (Table S1). We next checked whether the sequences matched mtDNAs from different types of humans (modern humans, Neanderthals, and Denisovans). Seven of the 8 libraries contained DNA fragments carrying Denisovan-derived nucleotide variants, whereas none of the libraries showed Neanderthal-derived variants (Table S1). These 7 libraries were used in further analyses.



Figure 2. The position sampled for the 0.3 mg of dental calculus on the Harbin tooth, where mtDNA was captured and sequenced

Contamination estimates

To enrich for putatively endogenous DNA, we selected fragments with a C to T substitution relative to the modern human reference at one end and checked the C to T substitution frequencies at the opposite end of such fragments. A substantially higher frequency at the opposite end relative to fragments without a C to T substitution at the initial end indicates that a substantial proportion of the total fragments derive from present-day DNA contamination. For the Harbin individual, the C to T substitution frequency at the 5′ ends for fragments with C to T substitution at the 3′ ends was 34% relative to 14% (Figure 3A, orange) for all fragments. Conversely, at the 5′ ends it was 52% relative to 20% (Figure 3A, blue, and Figure S2). This indicates that many of the mtDNA fragments in the libraries represent contamination from present-day humans. Furthermore, the C to T substitution frequencies indicates that endogenous DNA was primarily present among fragments shorter than 60 bp (C to T% >10%) (Figures 3B and S1).

Two approaches to estimate contamination levels from deamination patterns[39,40] (STAR Methods) yielded estimates of 56% and 67%, respectively. We also find that 82% (225/276) of fragments that cover positions specific to modern humans carry modern human variants. We note, however, that 15% (95/616) of fragments that cover variants specific to Denisovans carry Denisovan variants, and 0% (0/241) carry Neanderthal-specific variants at positions unique to Neanderthals (Table S1).

DNA sequence reconstruction

To avoid biasing the mtDNA fragments identified by using a reference sequence from a certain group of hominins, we aligned the DNA fragments captured to a reconstructed human ancestral mtDNA inferred from the majority of 23 chimpanzees,[41] the Sima de los Huesos hominin, 18 Neanderthals,[15,29,42–46] 7 Denisovans, and the modern human Reconstructed Sapiens Reference Sequence (RSRS[47]) (Table S2, STAR Methods). To explore the relationship of the Harbin mtDNA to Denisovan mtDNAs, we maximized the recovery of additional endogenous mtDNA fragments from the Harbin dental calculus by using a reconstructed ancestral Denisovan mtDNA reference sequence of 7 Denisovans (Table S2, STAR Methods).

To enrich for endogenous ancient DNA fragments, we restricted our analysis to fragments shorter than 60 bp where the first and last three terminal nucleotides showed evidence of cytosine deamination ("deaminated," Table S2). By combining such mtDNA fragments from 7 libraries (Table S1), we obtained 828 unique fragments aligned to the reconstructed human ancestral mtDNA and 855 unique fragments aligned to the reconstructed ancestral Denisovan mtDNA sequence (Table S2). To further reduce errors introduced by deamination of cytosine residues, we masked positions with C→T mismatches in the forward strand and G→A in the reverse strand relative to the respective ancestral human or ancestral Denisovan reference sequence. The consensus bases for the mtDNA genome were then called using positions covered by two or more fragments where at least two-thirds of the bases agreed. This yielded a consensus mtDNA sequence covering 7,817 nucleotide positions for the sequences aligned to the reconstructed ancestral human mtDNA and 8,234 nucleotide positions for the sequences aligned to the reconstructed ancestral Denisovan mtDNA (Table S2).

Phylogenetic reconstruction and Denisovan mtDNA variation

We next estimated phylogenetic trees using a Bayesian framework,[48] maximum likelihood,[49] and maximum parsimony[50] for the Harbin mtDNA consensus sequences and the mtDNAs from the Sima de los Huesos hominin, 7 Denisovans, 18 Neanderthals, 58 present-day humans, and a chimpanzee.[41] All three trees share the same topology (Figures 4 and S4) where the Harbin mtDNA falls within the Denisovan mtDNAs. Within the Denisovan branch, the Harbin mtDNA tends to cluster with the older Denisovans (*Denisova 2*, *Denisova 8*, *Denisova 19*, *Denisova 20*, and *Denisova 21*) relative to the two younger Denisovans (*Denisova 3* and *Denisova 4*), albeit with low support (33% or 43% posterior, 44% or 39% bootstrap) (Figures 4 and S4).

To further explore the relationship of the Harbin mtDNA to the Denisovan mtDNAs, we then analyzed six positions covered in the Harbin mtDNA, where the five older Denisovans (*Denisova 19*, *Denisova 20*, *Denisovan 21*, *Denisova 8*, and *Denisovan 2*) were fixed for derived variants and the two younger Denisovans (*Denisova 3* and *Denisova 4*) carried ancestral variants (Table 1). Four of these were covered by two or three DNA fragments, whereas two were covered by single fragments. Of the six positions, five carry the derived variants seen in the older Denisovans, whereas one carries the ancestral variant (position 16,111, covered by 4 fragments) (Table 1). The derived variants of the two positions covered by single mtDNA fragments are A and C (Table 1) and thus cannot have been introduced by deamination.

We then analyzed the 25 or 24 positions covered in the Harbin mtDNA, where the two younger Denisovans (*Denisova 3* and *Denisova 4*) carry derived variants and the five older Denisovans have ancestral variants. All of these positions carry ancestral variants in the Harbin mtDNA. Thus, five positions support that the Harbin mtDNA is related to the mtDNA of the older Denisovans and no positions support a relationship to the mtDNAs of the younger Denisovans, although one position suggests that the Harbin mtDNA could be basal to the older Denisovan mtDNAs.

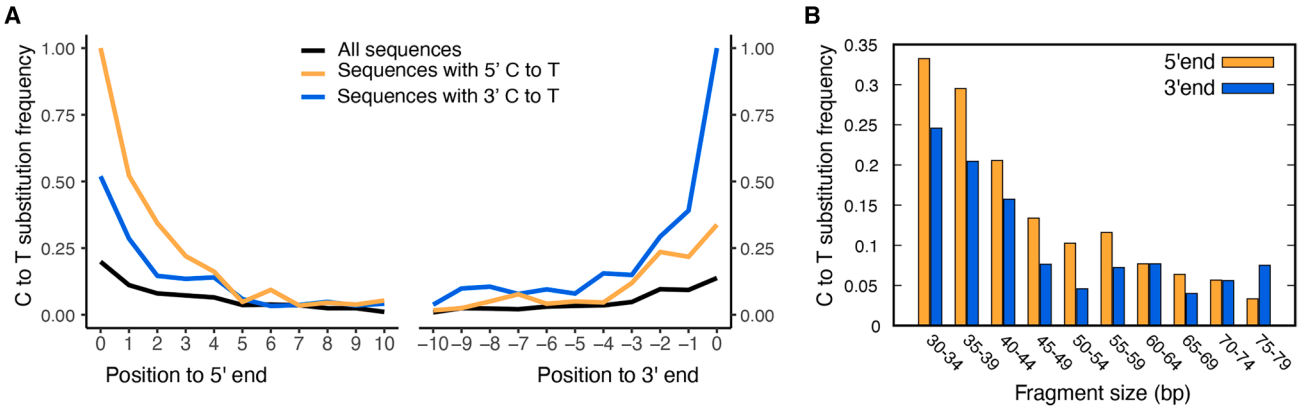


Figure 3. Cytosine deamination features of the Harbin mtDNA from captured sequences

(A) C to T substitution frequencies at the ends of reads for all sequences (black). The orange line represents selecting sequences carrying C to T substitutions at their 5′ ends, so the C to T substitution frequency is 100% at the 5′ ends and 34% at the 3′ ends. The blue line represents selecting sequences carrying C to T substitutions at their 3′ ends, where the C to T substitution frequency is 100% at the 3′ end and 52% at the 5′ end.

(B) C to T substitution frequencies at sequences' first and last ends in every 5-bp length bin.

Related to Figure S2.

DISCUSSION

Previous morphological studies[1,2,51] have classified the Harbin individual as a new species or a new sister lineage to *H. sapiens*. However, we show that the Harbin individual's mtDNA falls within the variation of previously sequenced Denisovan mtDNAs and is likely to be more closely related to the mtDNAs of the Denisovan individuals *Denisova 2*, *Denisova 8*, *Denisova 19*, *Denisova 20*, and *Denisova 21* than to *Denisova 3* and *Denisova 4*.

Denisova 19, *Denisova 20*, and *Denisova 21* range in age from 187 to 217 ka[13,31] and were found in the oldest archaeological layer of the East Chamber in Denisova Cave. Along with *Denisova 2* (~123–194 ka) and *Denisova 8* (~190 ka), the mtDNAs of these older Denisovans form a clade to the exclusion of the younger *Denisova 3* (~52–76 ka) and *Denisova 4* (~55–84 ka).[31] The Harbin individual is similar in age (cranium date: >146 ka; sediment date: 138–309 ka)[2] to the older Denisovans from Denisova Cave, and the Harbin mtDNA tends to be associated with the mtDNAs of these older individuals. This shows that Denisovans inhabited a large geographical range in Asia in the Middle Pleistocene.

The finding that the Harbin cranium contains Denisovan mtDNA connects Denisovans with morphological traits and will help to identify other specimens as Denisovans, especially when ancient DNA is not preserved or difficult to obtain. For example, fossils from Dali[52,53], Jinniushan,[52,53] and Hualongdong[3] exhibit morphological traits[1] similar to those in the Harbin cranium, making it plausible that they also represent Denisovan populations.

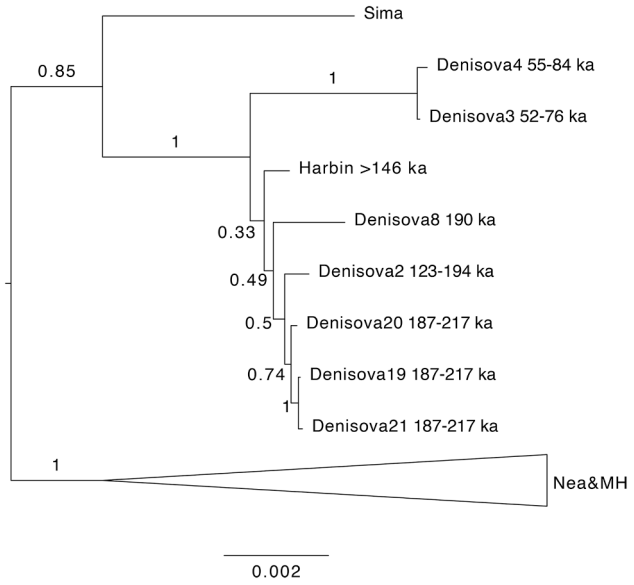


Figure 4. Phylogenetic subtree of hominin mitochondrial relationships using the Harbin mtDNA consensus sequences aligned to the reconstructed sequence of the common ancestor of Denisovans

The triangle in the tree collapses a clade of 18 Neanderthals and 58 modern humans. The tree was constructed with a Bayesian phylogeny estimated under a GTR+G model. Numbers on the tree indicate posterior probabilities for the clade. The scale bar denotes substitutions per site. The tree was rooted using a chimpanzee genome (GenBank accession number HM068593.1).

Related to Figure S4.

Limitations of the study

This study is based on the limited amount of hominin mtDNA that could be retrieved from the dental calculus of the Harbin cranium. To enhance our understanding of the genetic relationships of Middle Pleistocene hominins, data from nuclear DNA will need to be retrieved from fossils from across East and Southeast Asia. Nevertheless, the finding that the human DNA of the Harbin specimen is better preserved in the dental calculus than in dense bones, including the petrous bone, suggests that dental calculus may be a valuable source for investigating DNA in Middle Pleistocene hominins.

Table 1. Summary of the variants at positions derived in older Denisovans but ancestral in younger Denisovans

rCRS_position	Ancestral	Derived	Harbin allele	Coverage	Early Denisovans	Late Denisovans	Sima de los Huesos
1,393	A	G	G	2	GGGGG	AA	A
6,956	T	A	A	1	AAAAA	GG	T
9,380	A	G	G	2	GGGGG	AA	A
11,623	C	T	T	3	TTTTT	CC	C
16,111	C	T	C	4	TTTTT	CC	C
16,320	T	C	C	1	CCCCC	TT	T

Early Denisovans = *Denisova 2*, *Denisova 8*, *Denisova 19*, *Denisova 21*, and *Denisova 21*; Late Denisovans = Denisova *3* and *Denisova 4*; Harbin sequences were mapped to the common-ancestor-reconstructed sequence of 7 Denisovans.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to, and will be fulfilled by, the lead contact, Qiaomei Fu (fuqiaomei@ivpp.ac.cn).

Materials availability

This study did not generate new reagents.

Data and code availability

The raw fastq files of mtDNA-captured sequences are available through the Genome Sequence Archive[54] in the National Genomics Data Center[55] (https://ngdc.cncb.ac.cn/gsa-human/; accession no. HRA009322); the mitochondrial DNA sequences of Harbin, updated Sima and updated Denisova 2, and the two ancestral references (*FASTA* format) have been deposited in GenBase[56] in the National Genomics Data Center[55] (https://ngdc.cncb.ac.cn/genbase; accession no. C_AA108312-C_AA108317) and the Beijing Institute of Genomics, Chinese Academy of Sciences/China National Center. All software used is open access and available online. The script for extracting deaminated fragments and masking possible deamination bases along fragments is available at https://github.com/QFuLab/MaskDamageReads. For any additional information needed to reanalyze the data reported in this paper, please contact the lead contact.

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AUTHOR CONTRIBUTIONS

Q.F. designed the research project. Q.J. prepared the archaeological samples and materials. Q.F. and X.F. processed the data. Q.F. analyzed data. Q.F., P. C., and Q.D. performed or supervised wet laboratory work. Q.F. wrote the manuscript. Q.F., E.A.B., M.A.Y., W.P., and S.P. revised the manuscript. All authors discussed, critically revised, and approved the final version of the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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  - Contamination and basic filters
  - Alignment to different references, Ancestral sequence estimation, and Harbin consensus building
  - Assigning Denisovan-specific positions
- QUANTIFICATION AND STATISTICAL ANALYSIS
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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Biological Samples		
Dental calculus	This study	Harbin individual
Petrous and tooth	This study	Harbin individual
Chemicals, Peptides, and Recombinant Proteins		
Tween-20	Sigma	cat.no. P5927
1 M Tris-HCl, pH 8.0	AppliChem	A4577,0500
0.5 M EDTA, pH 8.0	AppliChem	A4892,1000
5 M NaCl	Sigma	S5150
25 mM each dNTP mix	Fermentas	cat.no. R1121
Bst DNA Polymerase, Large Fragment	NEB	cat.no. M0275S
ATP, 10 mM stock solution	NEB	cat.no. 9804
T4 Polynucleotide Kinase (10 U/μL)	NEB	cat.no. M0236L or S
T4 DNA Polymerase (3 U/μL)	NEB	cat.no. M0203L
Quick Ligation Kit	NEB	cat.no. M2200L
2x HI-RPM hybridization buffer	Agilent	5118-5380
20% SDS	Serva	39575.01
SSC buffer	Ambion	AM9770
AmpliTaq Gold 10x PCR buffer without Mg	Life Technologies	4379874
1M NaOH	Sigma	71463-1L
3M Sodium acetate pH 5.2	Sigma	S7899
Dynabeads MyOne C1	Life Technologies	65002
SeraMag Speedbeads	GE	65152105050250
Deposited Data		
Sequencing data are available in the BIG Data Center Genome Sequence Archive https://bigd.big.ac.cn/gsa-human Mitochondrial DNA FASTAs have been deposited in GenBase in the National Genomics Data Center https://ngdc.cncb.ac.cn/genbase	This paper	Genome Sequence Archive: HRA009322;GenBase mtDNA: C_AA108312-C_AA108317
Oligonucleotides		
Phosphate-AGATCGGAAG[C3Spacer] 10[TEG-biotin](TEG = triethylene glycol spacer)	Gansauge and Meyer[57]	CL78 Single-stranded adapter
GTGACTGGAGTTCAGACGTGTGCTCTTCC*GA*TC*T	Gansauge and Meyer[57]	CL130 Extension primer
CGACGCTCTTC-ddC (ddC = dideoxycytidine)	Gansauge and Meyer[57]	CL53 Double-stranded adapter, strand 1
Phosphate-GGAAGAGCGTCGTGTAGGGAAAGAG*T*G*T*A	Gansauge and Meyer[57]	CL73 Double- stranded adapter, strand 2
GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT-Phosphate	Fu et al.[38]	BO4.P7.part1.R
CAAGCAGAAGACGGCATACGAGAT-Phosphate	Fu et al.[38]	BO6.P7.part2.R
GTGTAGATCTCGGTGGTCGCCGTATCATT-Phosphate	Fu et al.[38]	BO8.P5.part1.R
GGAAGAGCGTCGTGTAGGGAAAGAGTGT-Phosphate	Yang et al.[58]	BO11.P5.part2.R
Software and Algorithms		
leeHom	Renaud et al.[59]	RRID: SCR_002710, https://github.com/grenaud/leeHom/
BWA 0.6.1	Li and Durbin[60]	RRID: SCR_010910, http://bio-bwa.sourceforge.net/

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Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
bam-rmdup 0.6.3	N/A	https://github.com/mpieva/biohazard-tools
mapDamage2	Hahn et al.[61]	https://github.com/ginolhac/mapDamage
contamMix 1.0-10	Fu et al.[62]	https://github.com/StanfordBioinformatics/DEFUNCT-env-modules/blob/master/contamMix/1.0-10
schmutzi	Renaud et al.[39]	https://github.com/grenaud/schmutzi
mafft	Rozewicki et al.[63]	http://mafft.cbrc.jp/alignment/software/
AuthentiCT	Peyregne and Peter[40]	https://github.com/Stephane Peyregne/AuthentiCT
jModeltest2	Darriba et al.[64]	https://github.com/ddarriba/jmodeltest2
Mrbayes (v3.2.6 x64)	Ronquist and Huelsenbeck[48]	https://nbisweden.github.io/MrBayes
RAxML-NG(v8.x)	Kozlov et al.[49]	https://github.com/amkozlov/raxml-ng
FigTree v1.4.4	N/A	http://tree.bio.ed.ac.uk/software/figtree
MEGA11	Tamura et al.[50]	https://www.megasoftware.net/download_form
MaskDamageReads	This study	https://github.com/QFuLab/MaskDamageReads

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Specimen description

A nearly complete hominin cranium was discovered in the Harbin area near Long Jiang (Dragon River) in Heilongjiang Province, China.[1,2] The cranium was determined to have a minimal age of 146 ka with direct U-series dating. Regional stratigraphic correlations based on the upper part of the Upper Huangshan Formation show a similar age of 138–309 ka.[65] Morphological analysis of the Harbin cranium suggests that the individual was from a hominin population that is a sister group to *H. sapiens*. The specimen was designated as a new species, *H. longi*.[2]

Ethics statement

Permission to test for ancient DNA in the human specimen from this study was obtained through discussions with the local researcher responsible for the Harbin specimen, and final approval was granted by Hebei GEO University, where the human specimen is managed. The work was done in collaboration with local researchers, who are co-authors due to their contributions to the collation of archaeological materials and/or discussions that informed the study.

METHOD DETAILS

Sample DNA extraction, library preparation, and mtDNA capture

The surface of the tooth was not treated before sampling. The 0.5 mg and 0.3 mg samples of dental calculus taken from the tooth refer to a scraped loose piece and a small tight piece, respectively. Before DNA extraction, 0.5 mg and 0.3 mg of dental calculus were treated with UV irradiation in a cross-linker and then washed with 5% bleach,[37] followed by an 80% ethanol rinse to reduce possible surface contamination from present-day humans and microbes. During the initial screening, the small tight piece of dental calculus (0.3 mg) was manually treated with an extraction buffer composed of two reagents (EDTA and proteinase K) to release DNA from the sample, resulting in 1000 ul liquid.[66] Two aliquots (each 140 ul) out of the 1000 ul liquid were processed on robotic platforms for further steps to obtain two extracts. After this, the remaining 720 ul liquid from the 0.3 mg sample, along with the 0.5 mg scraped loose dental calculus piece, were manually processed into two extracts, allowing for a larger volume during the binding step of extraction[66] (Table S1). Six additional extracts were obtained from the surface powder of the tooth. Two re-extracts were made from the silica membrane from the same column of the first extract, while three re-extracts were made from the pellet of the 0.3 mg and 0.5 mg dental calculus samples. The pellet extraction from the 0.3 mg dental calculus sample was performed twice to maximize DNA recovery. The second re-extract from the 0.3 mg dental calculus pellet was treated with a refined silica-based DNA extraction method to recover shorter fragments.[67] A total of 20 libraries were generated using a single-stranded library preparation protocol without UDG treatment.[68] We additionally prepared 5 extracts of petrous bone and one extract of dentine powder sample (4.8-9.5 mg each) and converted these into 8 libraries (Table S1). To assess the efficiency of the single-stranded library preparation, a small quantity of a control oligonucleotide was added to each library during the library preparation.[67]

AccuPrime Pfx DNA polymerase (Life Technologies) was used to amplify each library for 35 cycles. In this amplification process, sample-specific barcodes were incorporated into the P7 and P5 library adaptors.[69] Then, the resulting PCR products were purified with the MinElute PCR purification kit (Qiagen, Hilden, Germany), and the library concentrations were measured with a NanoDrop 2000 spectrophotometer.

Two rounds of hybridization capture at 65 °C were conducted using a modified on-bead hybridization protocol[70,71] with a modern human mtDNA probe set.[38] After the captured libraries were pooled, heteroduplexes in these pools were eliminated through a one-cycle PCR reaction using Herculase II Fusion DNA polymerase (Agilent Technologies)[72] with primers IS5 and IS6.[73] Subsequently, the DNA concentration in the pooled libraries was evaluated on a Bioanalyzer using a DNA-1000 chip (Agilent Technologies).[72]

Sequencing, raw data processing, and mapping

The sequencing of the libraries was performed for 76 cycles from both ends of the library insert, incorporating two additional 7-base pair index reads through double-index sequencing[69] on Illumina MiSeq platforms (MS-102-3001 MiSeq Reagent Kit v3,150-cycle). After introducing the indexed control PhiX 174 library, the yield of control reads (index 5'-CGATTCG-3') was between 1.47% and 1.99%. After base calling with Bustard (Illumina), *leeHom*[59] was utilized to trim and merge the paired sequences with at least 11 overlapping base pairs. The downstream analyses focused exclusively on the fragments containing the expected indices for each library.

After being enriched with the human mtDNA probe set, the sequences were aligned to the revised Cambridge Reference Sequence (rCRS) (NC_012920.1,[74]) using *bwa* (version: 0.5.10-evan.10-1-gfb1ff83).[60] This process used the parameters: –n 0.01 –o 2 –l 16500, which increases the tolerance of mismatches and gaps and turns off seeding. After mapping, PCR duplicates were removed by calling a consensus from the sequences with identical alignment start and end coordinates using bam-*rmdup* (https://github.com/mpieva/biohazard-tools version: 0.6.3, Table S1). Sequences with a mapping quality lower than 25, a length shorter than 30 bp, or without a paired end were removed (Table S1).

Enrichment of mitochondrial sequences

The efficiency of human mtDNA enrichment varied from 33.8% to 90.2% of the sequences $\geq$ 30 bp before duplicate removal (Table S1). The sequence duplicates ranged from 8 to 689. However, for powder excluding material from the surface of the tooth, the sequence duplicates of the libraries from the purely dental calculus and bone samples ranged from 23 to 685 (Table S1), indicating that the fragments in these libraries were exhaustively sequenced. The unique sequence counts (Table S1) were 139 and 156 for the libraries from re-extracted bone pellets, 242 to 568 for the libraries from re-extracted silica membrane, 487 to 3081 for the dental calculus sample libraries, and 659 to 1138 for the bone sample libraries. For the terminal C to T substitution frequencies in libraries, 5 libraries of the 0.3 mg sample ranged from 16.1-31.1%, which is higher than 10%; in contrast, the libraries from the 0.5 mg dental calculus sample ranged from 1.0% to 3.1%, and the libraries from bone samples ranged from 0% to 5.6%. As ancient DNA possesses an elevated frequency of terminal C to T substitutions,[75,76] most libraries from the 0.3 mg dental calculus sample were concluded to contain authentic ancient hominin DNA fragments. 5-10% terminal C to T substitutions were present in three libraries of the 0.3 mg dental calculus sample and one library from the tooth. Based on later analyses, we know that the mtDNA from the Harbin individual is related to Denisovans, so here we assigned all fragments of these two libraries from the dental calculus and one library from the tooth to see how many of them covered Denisovan-specific positions. Two (V8492 and V8494) out of these 4 libraries with 5-10% C to T substitution frequencies were found to contain Denisovan-derived alleles. No Denisovan-derived alleles were found in libraries from the bone samples. Together with 5 libraries with high C to T substitution frequencies, 7 libraries from the 0.3 mg dental calculus sample were used for further analyses (Table S1).

Contamination and basic filters

A likelihood-based method (contamMix) was used to estimate contamination from present-day humans within the extracted mtDNA.[62] Contamination and errors were simultaneously evaluated through this likelihood-based method by comparing mtDNA fragments with a reference panel of 311 present-day humans worldwide, whose mtDNA haplotypes are treated as potential sources of contamination. The 7 libraries from the 0.3 mg dental calculus sample showed present-day human DNA contamination ranging between 0.5% and 22.2%. These libraries were combined, and the program schmutzi[39] was applied to estimate the contamination rate by using sequence deamination patterns. This showed a high present-day human contamination level of 57%. Additionally, using the AuthentiCT program[40] with an ancient DNA damage model, the Harbin merged sequences were found to contain 67% present-day DNA contamination.

Since the contamination estimates determined using contamMix, schmutzi, and AuthentiCT were different, we additionally implemented a contamination method specifically for archaic humans: estimating the proportion of modern human-derived alleles in the modern humans' mtDNA diagnostic positions. Here, we pinpointed unique variants specific to modern humans, Neanderthals, and Denisovans by analyzing data from 1,058 modern humans, 18 Neanderthals (all with fewer than 5 missing positions), 7 Denisovans, a Sima de los Huesos individual, and 23 chimpanzees.[13,27,41,46,71,77] To distinguish different branches, we required that 99% of the positions of the 1,058 modern humans, 100% of the positions of the 18 Neanderthals, and 100% of the positions of the 7 Denisovans must be the same within their respective branches. Subsequently, the base pairs present in the DNA fragments from the combined 7 libraries of the Harbin individual at these informative positions were identified.

To eliminate contamination from present-day humans, the deaminated fragments were filtered by selecting those with C→T substitutions that occurred in any of the first or last three positions with MaskDamageReads (https://github.com/QFuLab/MaskDamageReads). To minimize the error caused by C to T substitutions in the deaminated fragments while building the consensus, Ts were masked as Ns in the forward strands of the DNA fragments, and As were masked as Ns in the reverse strands with MaskDamageReads.

Alignment to different references, Ancestral sequence estimation, and Harbin consensus building

To minimize alignment bias to the rCRS mtDNA sequence and maximize the recovery of endogenous sequences due to the divergence between human reference mtDNA and Denisovan mtDNA, sequences were aligned with the same alignment parameters as rCRS to two references (Table S2): i) a reconstructed putative ancestral human mtDNA sequence; ii) a reconstructed common ancestor sequence of 7 Denisovans.

The reconstructed putative sequence for 'Ancestral Humans' was based on the allelic patterns for the aligned mtDNAs of the alleles carried by the majority of 23 chimpanzees,[41] the Sima individual, 7 Denisovans, 18 Neanderthals, and the modern human Reconstructed Sapiens Reference Sequence (RSRS[47]). The Python script implemented the following steps:

1. For all positions where all 7 Denisovans had the same allele as RSRS and the Neanderthals, that allele was used for the reconstructed sequence (column 1, Figure S3i).
2. The putative ancestral allele for Neanderthals was determined based on one of the following criteria: If all 18 Neanderthals shared the same allele, that allele was chosen to be the Neanderthal ancestral allele. If they did not, but the most divergent Neanderthal (Hohlenstein-Stadel) shared the same allele as the modern human RSRS allele (Figures S3 and S3iii), or as the five early Denisovans or Sima (Figure S3ii), then that allele was chosen as the Neanderthal ancestral allele.
3. For positions where the 7 Denisovans varied and the Neanderthal and modern human alleles differed from each other (Figure S3ii), if all five early Denisovans shared the same allele and it was identical to either the Neanderthals or RSRS, that allele was designated as ancestral (column 3).
4. If the five early Denisovans differed, then the allele in Sima was used to determine the ancestral allele if it was identical to either the Neanderthals or RSRS (column 4).
5. For all other positions where the 7 Denisovans varied but all Neanderthals and RSRS matched each other, three patterns were used to determine the putative ancestral alleles (Figure S3iii): (a) If the Neanderthals and RSRS matched the chimpanzee (column 5), (b) the five early Denisovans (column 6), or (c) the Sima allele (column 7), this allele was regarded as ancestral. (d) If the Neanderthal and RSRS alleles did not match the chimpanzee allele but were shared with at least five early Denisovans (column 8), then that allele was regarded as ancestral.
6. For all other patterns not described above, the ancestral allele was not designated (labeled 'N').

For the ancestral Denisovan sequence ('Ancestral 7Dens'), at positions where all 7 Denisovans shared the same allele, we regarded this allele to be the ancestral allele, without taking into account the Neanderthal or modern human alleles (column 2, Figure S3i). When the 7 Denisovans did not share the same allele, we followed the procedure as for the 'Ancestral Humans' reconstructed sequence as described above (Figures S3ii and S3iii).

The deaminated fragments shorter than 60 bp aligned to each reconstructed reference were selected using MaskDamageRead, choosing those with C→T substitutions on the forward strand and G→A on the reverse strand (relative to the reconstructed reference) in any of the first or last three positions. To minimize errors caused by deamination during construction of the consensus, Ts were masked as Ns in the forward strands of the DNA fragments, and As were masked as Ns in the reverse strands. In all cases, we required that the agreement of the bases exceed two-thirds (>67%) of the overlapping fragments, and that each position was covered by at least two fragments.

Assigning Denisovan-specific positions

This analysis used data from 1,058 present-day humans, 18 Neanderthals, as well as mtDNA sequences from 7 Denisovans (*Denisova 2, Denisova 3, Denisova 4, Denisova 8, Denisova 19, Denisova 20, and Denisova 21*) from Denisova Cave and a Middle Pleistocene individual from Sima de los Huesos to identify positions specific to all 7 Denisovans (Table S3). To reduce incorrect identification, we required that there be no missing data from the 7 Denisovans and Sima de los Huesos for the positions specific to Denisovans.

QUANTIFICATION AND STATISTICAL ANALYSIS

Phylogenetic tree construction

MAFFT[63] was used to generate two separate sequence alignments for the Harbin consensus mtDNA sequences aligned to the human reconstructed ancestral mtDNA and the reconstructed ancestral mtDNA sequence estimated from the 7 Denisovans against 58 out of 1,058 present-day humans from the HGDP,[77] 18 Neanderthals, 7 Denisovans from Denisova Cave (*Denisova 2, Denisova 3, Denisova 4, Denisova 8, Denisova 19, Denisova 20, and Denisova 21*), and a Middle Pleistocene individual from Sima de los

Huesos.[13,27,46,71] The number of missing bases for these published ancient mtDNA sequences is listed in Table S4. A portion of the missing positions in the mtDNAs of *Denisova 2* (9 out of 11 positions) and Sima de los Huesos (118 out of 264) was recovered by aligning their sequences to the common ancestor reconstructed sequence of 7 Denisovans and the common ancestor reconstructed sequence of 7 Denisovans and Sima de los Huesos, respectively. All positions containing gaps or missing data were excluded from the phylogenetic tree analyses.

For the Bayesian statistical framework, MrBayes software[48] was utilized with a General Time Reversible sequence evolution model with a Gamma distribution model (GTR+G), executing 10,000,000 iterations of the Markov Chain Monte Carlo with the initial 1,000,000 iterations discarded as burn-in (Figures 4 and S4E). For the maximum likelihood (ML) criterion, RAxML-NG[49] was used with 500 bootstrap replicates under the GTR+G model identified through the best-fit model approach of jModeltest2[64] (Figures S4A and S4C). For the maximum parsimony (MP) method, MEGA 11[50] was applied using the subtree Pruning Regrafting (SPR) algorithm (Figures S4B and S4D) with 1000 random initial trees. All trees were plotted with FigTree v1.4.4 (http://tree.bio.ed.ac.uk/software/figtree) and were re-rooted with a chimpanzee genome.

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Supplemental figures

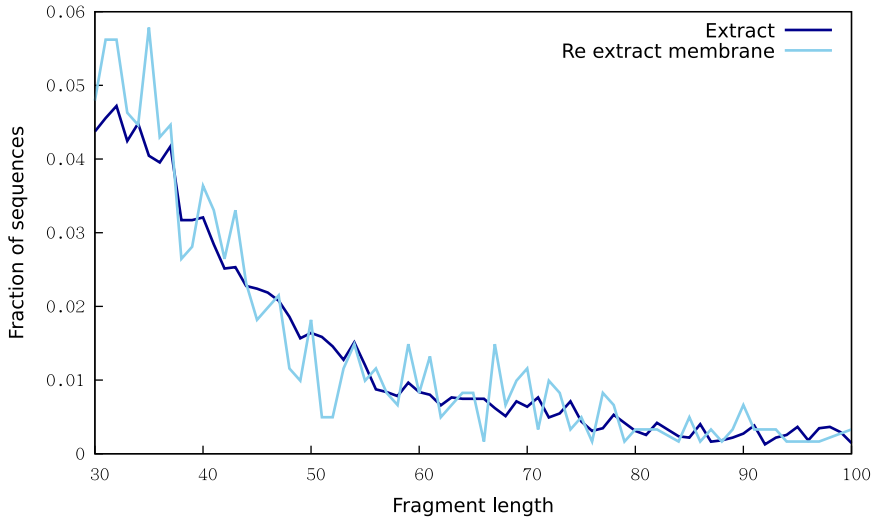


Figure S1. Sequence length distribution of unique sequences from 5 merged mtDNA-captured libraries of original extracts (dark blue) and 2 merged libraries of re-extracted silica membrane, related to STAR Methods

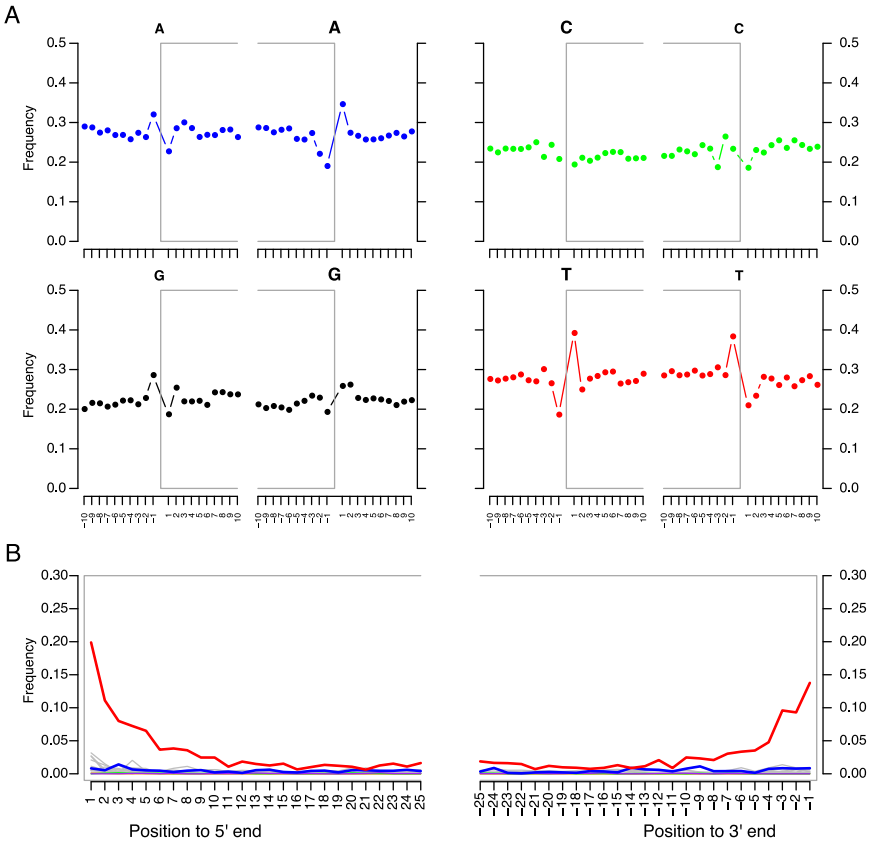


Figure S2. Deamination features of Harbin-captured mtDNA for all sequences, related to Figure 3

(A) Nucleotide frequencies of the reads (within gray boxes) and the surrounding reference sequence.

(B) Nucleotide misincorporation patterns around the beginning and ends of the mtDNA sequences. Red, C to T frequency.

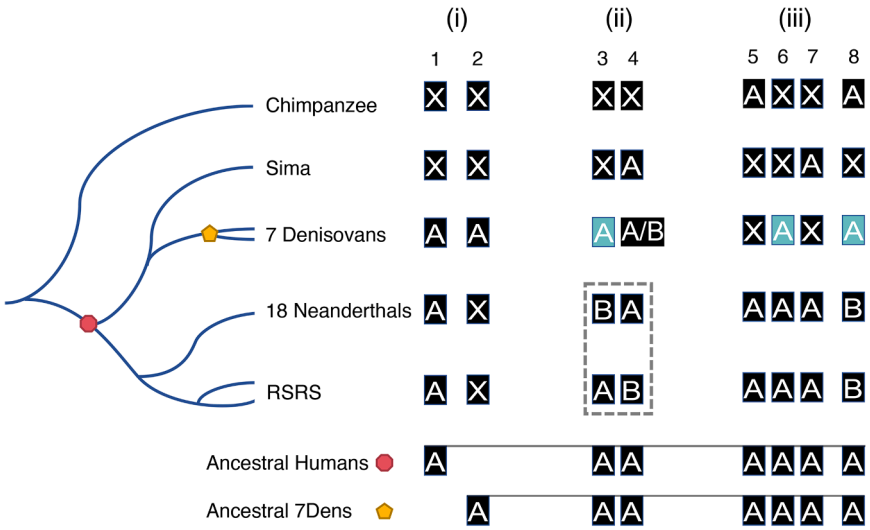


Figure S3. Schematic diagram of the method used to generate the sequences of the common ancestor of Denisovans, Neanderthals, and modern humans ("Ancestral Humans," red node) and the common ancestor of Denisovans ("Ancestral 7Dens," yellow node), related to STAR Methods

The tree includes mtDNAs from the majority of 23 chimpanzees, the Sima individual, 7 Denisovans, 18 Neanderthals, and the modern human Reconstructed Sapiens Reference Sequence (RSRS[47]). Boxes indicate the allelic pattern at an aligned position that was used for determining the ancestral allele for each reconstructed sequence, where A and B indicate the two possible alleles and X indicates that data from that group were not used in the ancestral allele determination for that pattern. See explanation in the STAR Methods section on "ancestral sequence estimation." Black shaded boxes for Denisovans indicate that all 7 Denisovans shared the same allele, whereas light blue shaded boxes indicate that at least 5 early Denisovans shared the same allele. The dashed line indicates an allelic pattern where the Neanderthal and modern human alleles do not match each other.

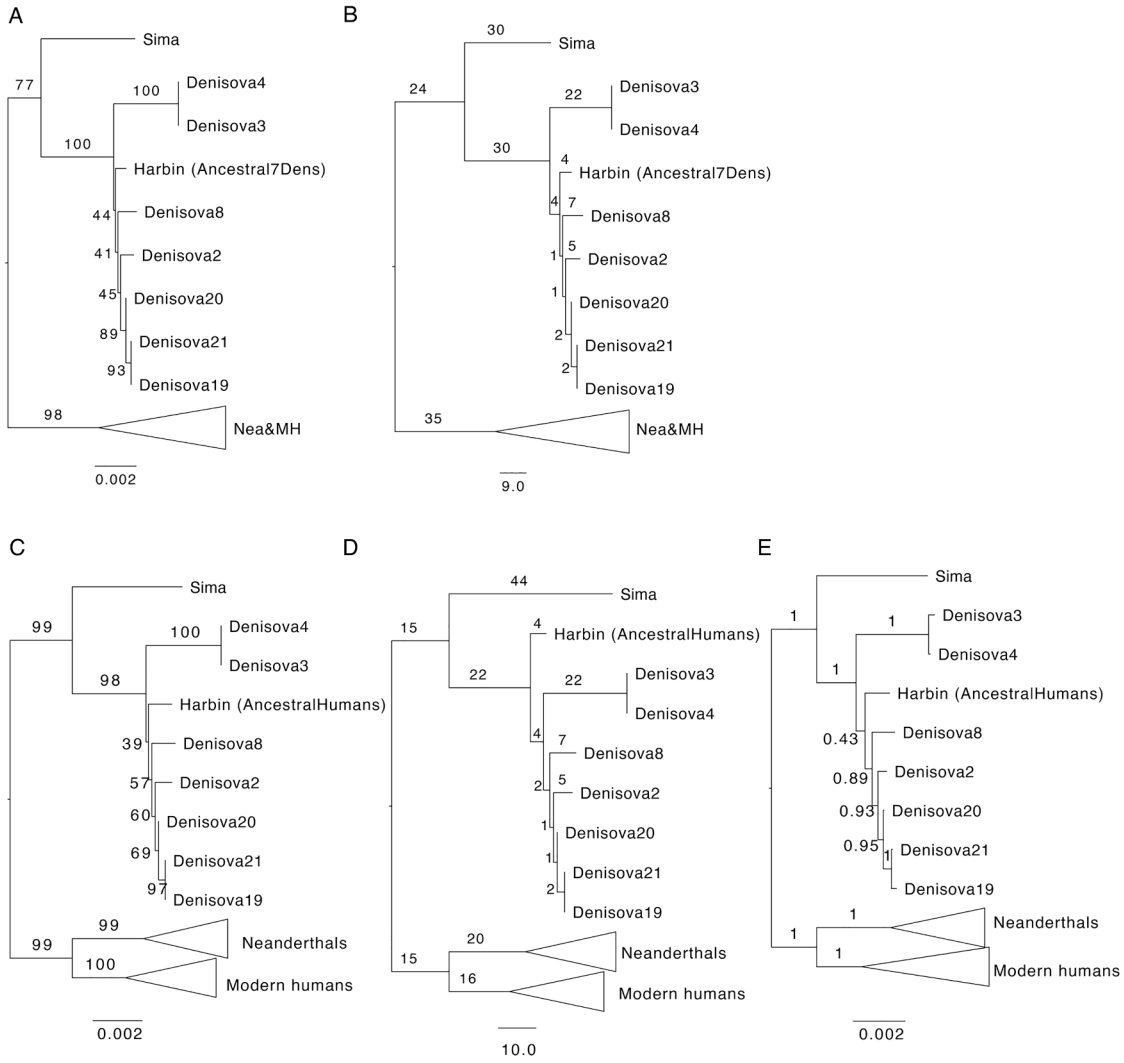


Figure S4. Phylogenetic subtree of hominin mitochondrial relationships using the Harbin mtDNA consensus sequences, identified with the reconstructed common ancestor sequence of 7 Denisovans (Ancestral7Dens) and the reconstructed human ancestral mtDNA sequence of all humans (Ancestral Humans), related to Figure 4

(A) A maximum likelihood tree for Harbin (Ancestral7Dens) estimated using a GTR+G model. Numbers on the tree indicate bootstrap values for the clade.

(B) A maximum parsimony tree for Harbin (Ancestral7Dens) estimated using the SPR model. The numbers on the branches represent the inferred number of substitutions.

(C) A maximum likelihood tree for Harbin (Ancestral Humans) reconstructed using a GTR+G model. Numbers indicate bootstrap values for the clade.

(D) A maximum parsimony tree for Harbin (Ancestral Humans) was inferred with an SPR model. The inferred number of substitutions is given on the branches.

(E) A Bayesian phylogeny for Harbin (Ancestral Humans) was estimated under a GTR+G model, with the numbers on the tree indicating posterior probabilities for the clade and the scale bar representing substitutions per site. The triangles in the trees represent the collapse of the clades of 18 Neanderthals and 58 modern humans. The trees were rooted using a chimpanzee mtDNA sequence (NCBI accession number HM068593.1).